

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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20. Perform Sharpening of Image using Laplacian mask with negative center coefficient.

0	1	0
1	-4	1
0	1	0

AIM:

Sharpening of Image using Laplacian mask with negative center coefficient.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png')

if img is None:
    print("Error: Image not found!")
else:
    # Sharpening kernel
    kernel = np.array([
        [0, 1, 0],
        [1, -4, 1],
        [0, 1, 0]
    ])

    # Apply the sharpening filter
    sharpened = cv2.filter2D(img, -1, kernel)

    # Save the output image
    cv2.imwrite(
        r'C:\Users\Susmitha\Downloads\ITA0510 Lab\Sharpened_Image.jpg',
        sharpened
    )
```

```
# Display the images
cv2.imshow("Original Image", img)
cv2.imshow("Sharpened Image", sharpened)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.